The Verification Boundary: institutional independence as a limit on platform expansion in AI markets

Summary brief · Anand Arivukkarasu · Supply Chain of Intelligence · Version 1.1 · July 2026 · supplychainofai.com/papers · Companion: Academic Theory Paper v4.1 (15 pp.), which is the canonical statement.

ABSTRACT

AI platforms have strong incentives to absorb adjacent workflow, data, and assurance functions. Existing theories explain why integration occurs; none isolates a recurring case in which integration destroys the value of the asset being acquired. This work develops the **verification boundary**. Where an automated output can impose material, difficult-to-recover loss beyond the producer, where a third party conditions money, permission, coverage, admissibility, authorization, or liability on an assurance about that output, and where conformity can be assessed against a sufficiently determinate standard, accepted verification requires a governance arrangement that preserves institutional independence from generation. The scarce asset is not superior intelligence, nor even superior error detection. It is credible independence.

THE PROPOSITION

> *When an automated output can impose material, difficult-to-recover loss on a party beyond the producer; when a third party conditions money, permission, coverage, admissibility, authorization, or liability on an assurance about that output; and when conformity can be assessed against a sufficiently determinate standard accepted by the conditioning party, accepted verification will require a governance arrangement that preserves institutional independence from the generating function. Improvements in generation or checking capability may reduce the cost and scope of verification, but do not by themselves eliminate the independence requirement.*

SCOPE CONDITIONS (CONJUNCTIVE) AND FORMS OF INDEPENDENCE

Scope condition	Coded as
1. External consequence	An erroneous output can impose material, difficult-to-recover loss on a party beyond the producer. Coded ex ante from reversibility, litigation exposure, and insurance loss experience.
2. External conditioning	A third party conditions money, permission, coverage, admissibility, authorization, or liability on an assurance about the output.
3. Determinate standard	A sufficiently determinate standard, recognized by the conditioning party, exists against which conformity can be asserted. A quality preference does not qualify.

Independence admits four governance forms, in ascending strength: protected internal function; ring-fenced entity; regulated professional role; separate firm. **The boundary constrains governance control, not necessarily firm boundaries** — separate-firm formation is a contingent response, not the proposition.

POSITIONING

Existing theories identify technical, regulatory, transaction-cost, legitimacy, and capability constraints on integration; none isolates the condition examined here, where controlling the assurance function destroys the third-party credibility that gives it value. Nearest neighbours: platform envelopment (Eisenmann, Parker & Van Alstyne, 2011); appropriability (Teece, 1986); transaction costs (Coase, 1937; Williamson, 1985); credence goods and audit institutions (Akerlof, 1970; Dulleck & Kerschbamer, 2006; DeAngelo, 1981; Power, 1997). The full paper also examines independent V&V (Lewis, 1992), conformity assessment (Terlaak & King, 2006; Büthe & Mattli, 2011), structural separation (Newbery, 1999), and credible commitment (North & Weingast, 1989), any of which may already contain the claim. Current anchors: SR 11-7 independent model validation; the mixed self-assessment and notified-body regime of the EU AI Act. Within the Supply Chain of Intelligence taxonomy the boundary grounds Layer 3, Gatekeeping; the taxonomy is context, not a premise.

Falsifiable predictions and refutation standard

Five-year horizon from publication. P1–P3 test the mechanism; P4–P6 test the claim against its two closest rivals, the regulatory account and the transitional account.

	Expected observation	Refuted if
P1 Separation	In markets meeting all three conditions, accepted assurance flows through a protected function, ring-fenced entity, regulated role, or separate firm.	Conditioning parties routinely accept generator assurance with no governance separation, across multiple cycles.
P2 Capability	As accuracy rises, verification automates and narrows; independent accountability persists.	Higher accuracy systematically causes assurance requirements to be dropped.
P3 Expansion bends	Platforms expand into workflow, access, and surface more readily than into assurance of their own outputs.	A platform repeatedly absorbs such assurance, controls its governance, and keeps third-party acceptance.
P4 Private demand	Separation appears wherever a private payer conditions money or liability, absent any mandate: at least half the rate observed in matched regulated markets.	Separation appears only under statute, and unregulated consequential markets integrate assurance without buyer resistance.
P5 Acquisition	Acquired assurance providers are ring-fenced or lose third-party acceptance.	Fully integrated providers keep acceptance and pricing power over multiple cycles.
P6 New mandates	Within 24 months of a new requirement, an accredited or separately governed provider is accepted in a majority of observed jurisdictions.	Newly regulated domains resolve into single-vendor generation-plus-verification.

WHOLE-CLAIM REFUTATION, WITH CODING RULES FIXED IN ADVANCE

One well-documented market in which all three scope conditions hold and the generator's own attestation is **durably accepted** by the conditioning third party. Durably accepted: reliance continues across more than two renewal, audit, or certification cycles. Scope conditions: coded from consequence and conditioning measures independently of whether separation is observed. No governance separation: the attesting unit has no protected reporting line, no non-overridable refusal right, and no external accreditation. A case meeting these rules settles the question.

CONCEDED LIMITS

The claim does not hold that separation guarantees good verification: issuer-pays ratings and the Andersen-era audit failures show nominal independence can be hollowed out, and the proposition then predicts crisis and re-separation rather than permanent absorption. Where an assertion reduces to *this computation ran on this input*, cryptographic attestation can displace institutional independence entirely; the claim narrows to assertions containing irreducible judgment. Vendor indemnification substitutes a balance sheet for an assertion, and is predicted to work only where the loss is financial and recoverable.

THE ONE QUESTION ASKED OF REVIEWERS

Is the verification boundary already established under another name? If it is, a citation is the most useful reply and the work should be withdrawn to a summary of prior results. If it is not, the sharpest contribution is a counter-example satisfying all three scope conditions that survives the coding rules above.

HOW TO CITE

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